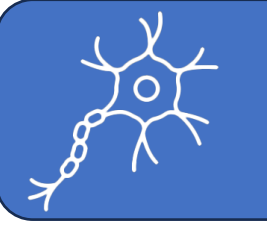




Seeking Criticality In Real Brain Networks



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Abstract

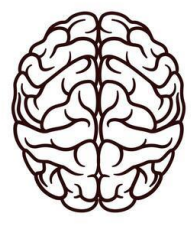
The **critical brain hypothesis** proposes that the brain operates near a critical point, where certain response functions diverge at a phase transition. Classically, reaching such a point requires precisely tuning an external control parameter (e.g., temperature) — raising the question of:

What could tune the brain to criticality or whether fine-tuning is even necessary?

Here, we explore the possibility that quenched structural heterogeneity in real neural networks relaxes this requirement: rather than a single sharp phase transition, disordered network topology may give rise to a **Griffiths phase** — *an extended region of parameter space exhibiting critical-like dynamics without the need for precise tuning* —. We investigate this framework using an excitable dynamics model on empirical connectome networks.



Terminology

 Connectome: A comprehensive map of neural connections	 Hub: A node with unusually high connectivity.
Degree	number of connections a node has
Heterogeneity	non-uniformity in node connectivity
Quenched heterogeneity	static disorder, not time-varying
Parameter space	the space of all possible control-parameter values



Data

The whole-brain connectome of *Drosophila melanogaster*:

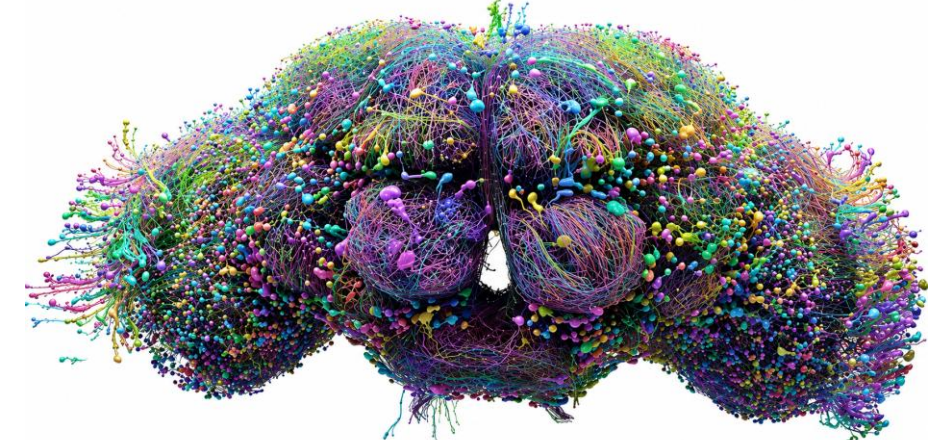


Fig. 1: Female *Drosophila Melanogaster* Full Brain[1]

- generated through **electron microscopy imaging**
- a **directed** and **weighted** network
- each **node** corresponds to a **neuron**
- **edge weights** : **total number of synapses**
- **between neuron pairs**

The network comprises 139,255 nodes, 3,732,460 edges, and 50,666,648 synapses.

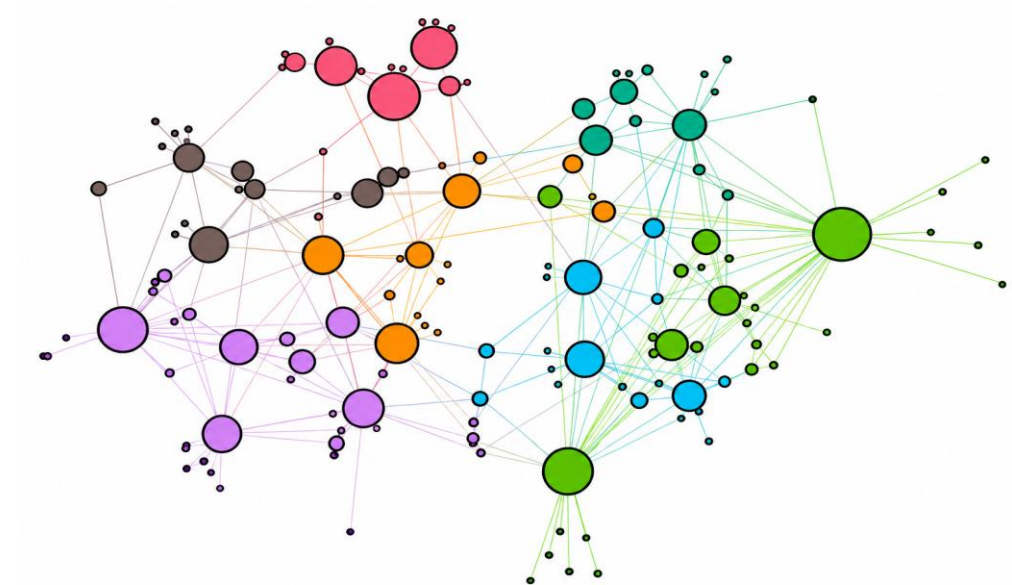


Fig. 2: Modular structure of the human connectome.[2]

- Each circle is a cluster of densely interconnected nodes.
- circle size reflects the number of nodes it contains.

The structural connectome of the *human brain*:

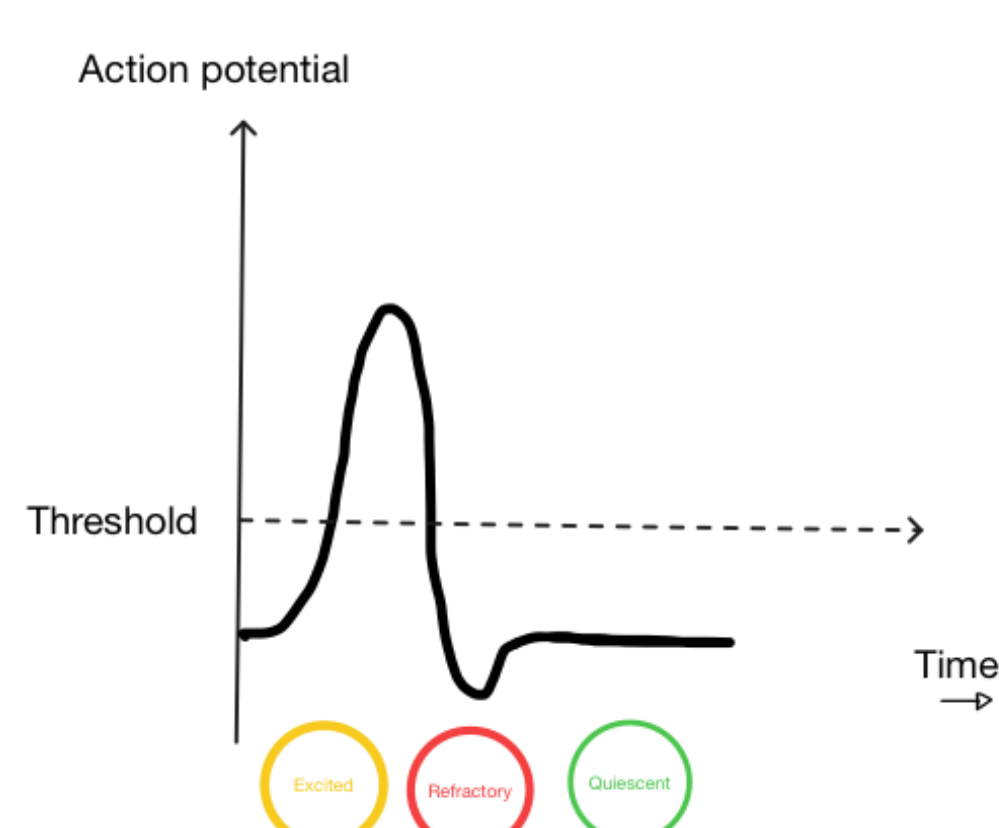
- generated through Diffusion Tensor Imaging (DTI) at 1 mm³ voxel resolution
- an **undirected** and **weighted** network
- each **node** corresponds to a **brain voxel**
- **edge weights**: **number of white matter fiber tracts** connecting each voxel pair

The network comprises 848,849 nodes and 66,298,600 edges.



Model

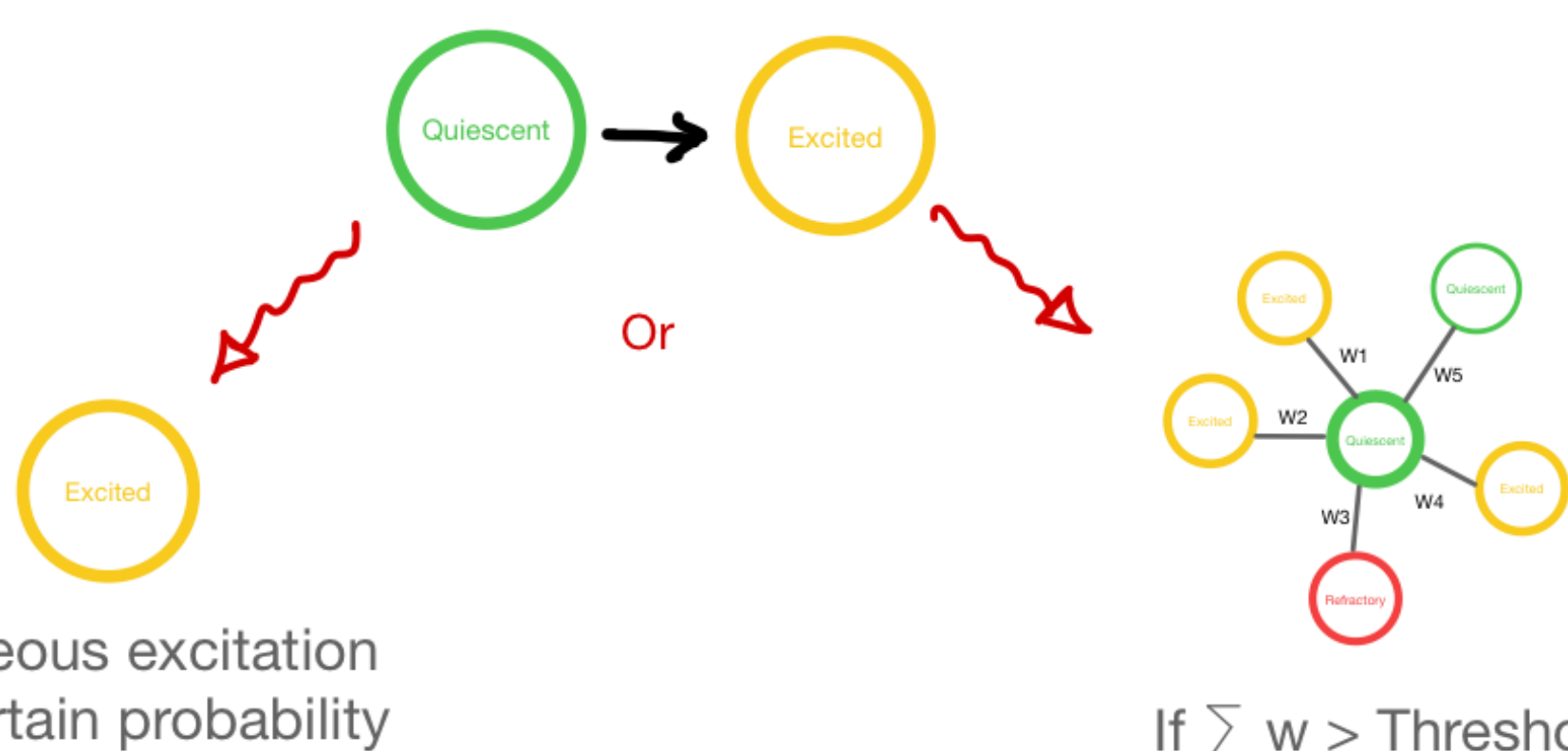
Recording neuron's activity:



Greenberg-Hastings model :

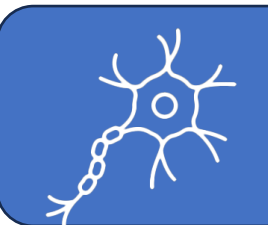
- A discrete model
- nodes are synchronously updated
- With certain transition rules
- three possible states

In this study, the formulation proposed by Haimovici et al. [3] is adopted.



What do we measure?

Size of “largest cluster” and “second largest cluster” formed by structurally connected excited neurons at each threshold (our control parameter)



Results

Size of largest cluster and second largest clusters measured for different thresholds:

- The first diagram: whole network
- The second diagram: 1% of high degree nodes removed from network

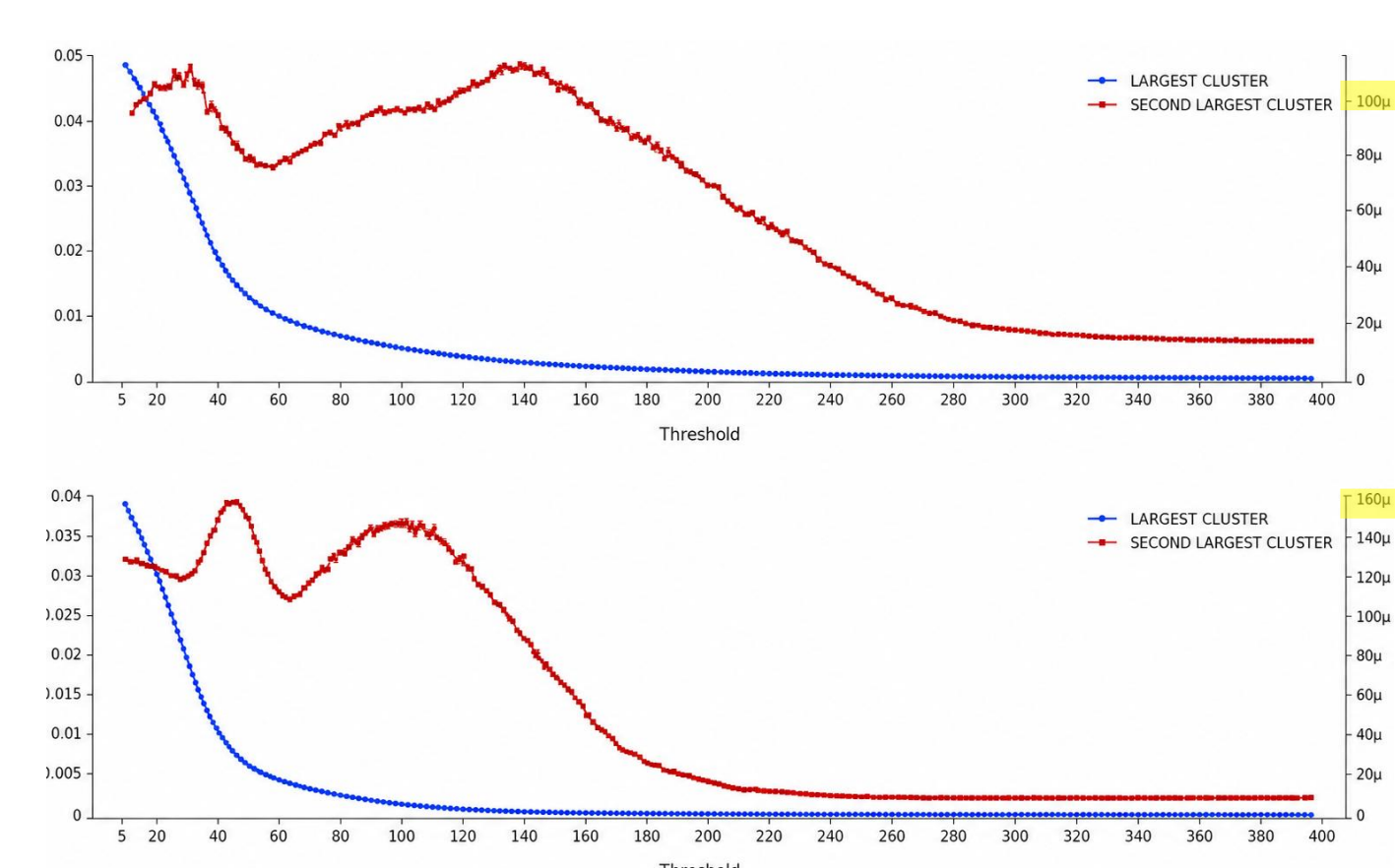
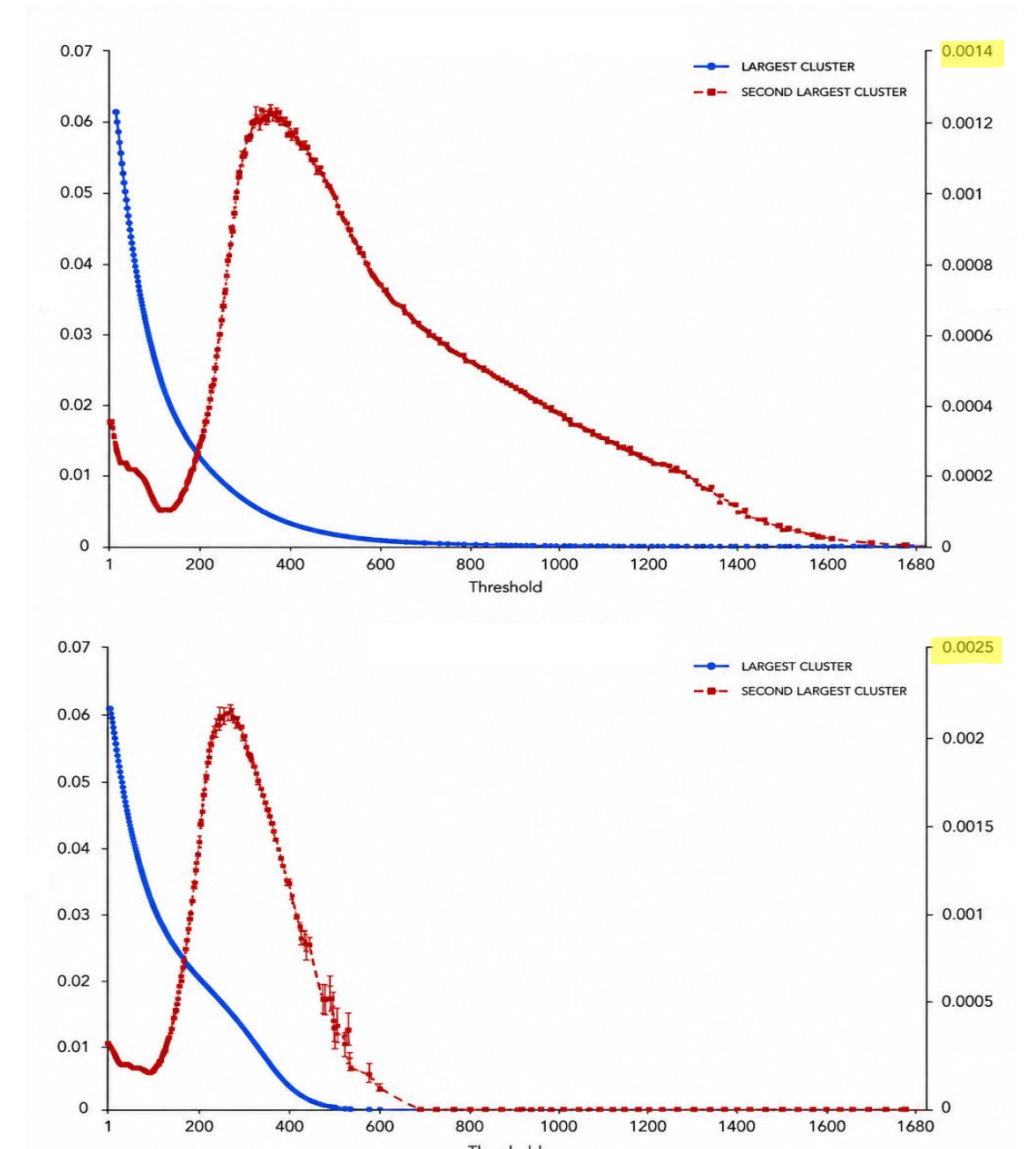


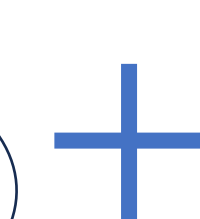
Fig. 3:
• *Drosophila connectome*

Fig. 4 :
• *human connectome*



Conclusion

The **increase in the peak height**



decrease in peak width
upon removing
a fraction of hub nodes



a **sharpening** of the transition
of second largest cluster curve

hub-driven structural heterogeneity underlies the smeared, **Griffiths-phase-like behavior** seen in the intact network.



Future Works



The emergence of **two distinct peaks** in the size of the second-largest cluster in the *Drosophila* brain network warrants further investigation.



References



Contact us

We would be delighted to hear your comments on this work!

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